Smart Safety Construction Helmet

Problem Statement

- Construction and mining sites are highly dangerous
- Traditional helmets only provide physical protection
- Accidents occur due to gas leakage, falls, and unsafe zones
- Smart helmet improves safety using sensors and IoT

Components Used

- ESP32 Microcontroller
- Gas Sensor (MQ-2 / MQ-135)
- Proximity Sensor (IR)
- MPU6050 (Accelerometer)
- Temperature Sensor (LM35)
- Humidity Sensor (DHT11)
- USB Power Module
- Safety Helmet

Working Principle

- Sensors continuously collect environmental data
- ESP32 processes sensor readings
- Unsafe conditions trigger alerts
- Data can be monitored using Wi-Fi

Power Supply

- ESP32 powered through USB
- Gas sensor powered using USB 5V module
- All components share a common ground
- Safe and stable power distribution

Advantages

- Improves worker safety
- Real-time monitoring
- Portable and wearable system
- Low cost and easy to implement

Applications

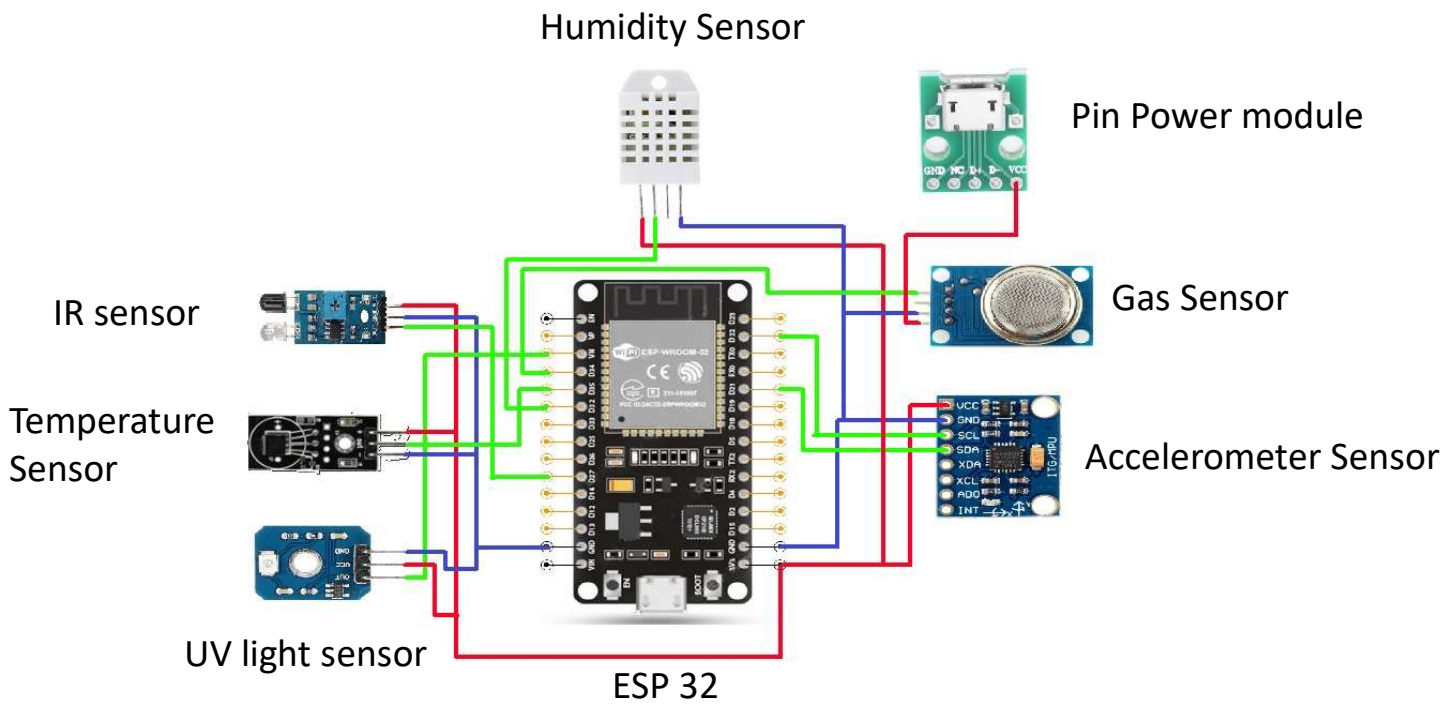
- Construction sites
- Mining industries
- Industrial safety monitoring
- Hazardous work environments

Cost Estimation

- **ESP32 (₹600–700) – Main controller; reads sensor data and sends alerts via Wi-Fi.**
- **Gas Sensor MQ-2 / MQ-135 (₹150–300) – Detects harmful gases in construction or mining areas.**
- **Proximity Sensor IR / Ultrasonic (₹100–200) – Detects unsafe distance from machines or danger zones.**
- **MPU6050 Accelerometer & Gyroscope (₹150–250) – Detects falls and abnormal head movement.**
- **Temperature Sensor LM35 / DS18B20 (₹50–100) – Monitors high surrounding temperature.**
- **Humidity Sensor DHT11 / DHT22 (₹100–200) – Detects high moisture or damp conditions.**
- **UV / Light Sensor (₹100–150) – Detects excessive sunlight (optional).**
- **Power bank(₹300–400) – Supplies portable power to the helmet system.**
- **Safety Helmet (₹300–500) – Base for mounting sensors and electronics.**

Circuit Diagram

Blue wires:- GND
Red wires:- 3.3V
Green wires:- Outputs



Conclusion

- Smart safety helmet enhances worker protection
- Uses sensors and IoT for real-time monitoring
- Reduces accidents and improves workplace safety

References

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